Multimedia Signal Processing - 2nd Module

June, 26 2020

1. **Question (8 pts)**
   Consider a discrete source X that emits as symbols the musical notes from the alphabet $[C, D, E, F, G, A, B]$. Consider the problem of encoding in a lossless fashion the song "Twinkle Twinkle Little Star", which can be represented by the sequence

   $$x = \{C, C, G, G, A, A, G, G, F, F, E, E, D, D, C, C\}.$$

   (a) Estimate the pmf of the source by considering the sequence x as a representative example.

   (b) Compute the entropy $H(X)$ of the source X.

   (c) Derive the Huffman code for the sequence x, showing a picture of the Huffman tree.

   (d) Compute the average code length of the Huffman code.

   (e) Encode the sequence x with the derived Huffman code (i.e., write the extension C^*).

   (f) Consider the case of encoding the same sequence by taking groups of two symbols at a time. Without doing any math, quickly explain how do you think the situation would change, and explain if you expect to gain something.

2. **Question (7 pts)**
   Oral examination.

3. **Questions (8 pts - each answer can be either TRUE or FALSE - Incorrect answers give you -1 points, unanswered 0 points, correct answers 1 point - Provide a brief comment when in doubt)**

   - [T] [F] Lossy coding schemes can provide less distortion than lossless coding ones at the same bitrate.
   - [T] [F] Consider scalar quantization of a source emitting uniformly distributed symbols within a closed interval. Lloyd-Max Quantizer and the Optimum Uniform Quantizer should converge to the same solution if the same amount of quantization steps is used.
   - [T] [F] In a video coding scheme, motion vectors can be estimated using different algorithms (e.g., diamond search, etc.) that always provide the same results.
   - [T] [F] In a vocoder, voiced speech signals can be well approximated using an all-poles filter fed with an input pitched signal.

- [T] [F] Consider an audio coding scheme using perceptual coding techniques. It is a good idea to consider the effect of both tones masking noise, and noise masking tones to decrease the coding bitrate achieving the same perceptual quality.
- [T] [F] Consider two audio tracks encoded using a perceptual coding scheme. The two audio tracks show two different mean square errors (MSE). The track with smaller MSE has the higher audio perceptual quality.
- [T] [F] You are developing a system that encodes a live music stream making use of transform coding based on KLT. Estimating the KLT basis from time to time might be preferable than estimating the KLT basis just once at the beginning of the stream.
- [T] [F] A closed-loop DPCM coding scheme may introduce more distortion than a simpler PCM scheme at the same bitrate.

4. **Question (Matlab - 10pts)**

   Consider a transform coding image encoding scheme. Load the standard RGB image 'peppers.png' running the command `img = imread('peppers.png');` (alternatively, create a random image by typing `img = uint8(rand(384, 512, 3)*255);`).

   (a) Display the image in a figure.

   (b) Extract and display the red component of the image and consider this as your signal from now on.

   (c) Encode the red image component using 2D DCT of size 4x4 and quantization step equal to 10.

   (d) Compute the minimum amount of bits needed to store this encoded image on disk (hint: only consider the signal, and forget about header information).

   (e) Decode the encoded image.

   (f) Display the decoded image.

   (g) Compute the introduced distortion in terms of PSNR.